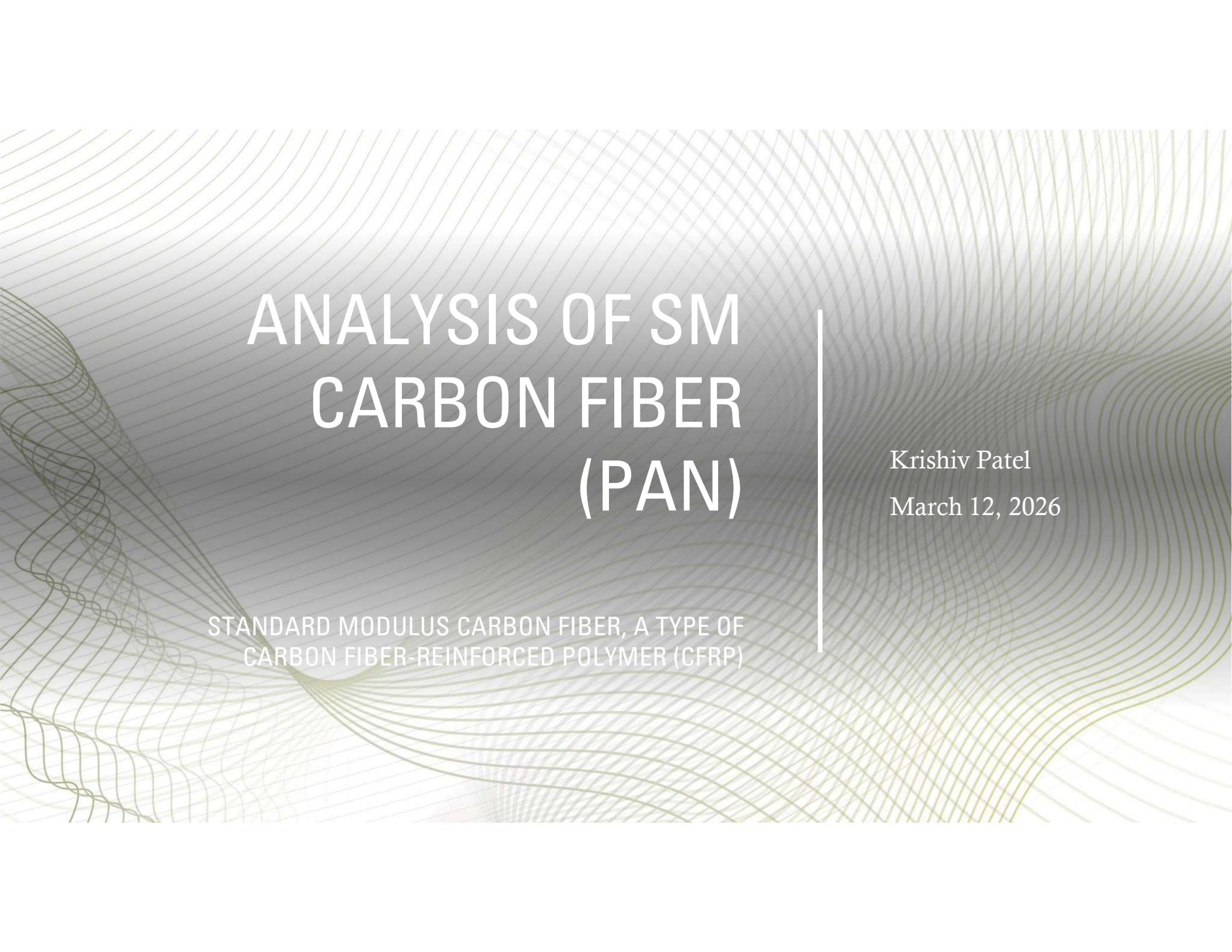




ANALYSIS OF SM CARBON FIBER (PAN)

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STANDARD MODULUS CARBON FIBER, A TYPE OF
CARBON FIBER-REINFORCED POLYMER (CFRP)

OVERVIEW

- Deliverables[3]
 - Input Data [4]
 - Assumptions[5]
 - Algorithms of Analysis[6]
 - Results[7-15]
 - Conclusions[16]
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DELIVERABLES

- Results of stiffness predictions for the laminated material
 - Values of E_{xx} , E_{yy} , G_{xy} , ν_{xy} , ν_{yx}
 - Results of deformation for the laminated material package
 - Table of $\langle \epsilon_{xx} \rangle$, $\langle \epsilon_{yy} \rangle$, $\langle \gamma_{xy} \rangle$
 - Results of strains in individual layers
 - Table of ϵ_{11} , ϵ_{22} , γ_{12} for $k=1, \dots, K$
 - Results of stresses in individual layers
 - Table of σ_{11} , σ_{22} , τ_{12} for $k=1, \dots, K$
 - Charts as functions of angle Θ and min/max values for calculated outputs
-

INPUT DATA

- Angle..... $\Theta = 13.3^\circ$
- Normal Stress (x-plane)..... $\sigma_{xx} = 1,528 \text{ MPa}$
- Normal Stress (y-plane)..... $\sigma_{yy} = 268 \text{ MPa}$
- Shear Stress..... $\tau_{xy} = 625 \text{ MPa}$
- Ply Thickness..... $h = .0001\text{m}$ (for scaling)
- Layup..... $[+Q/-Q/+30/-30s]$

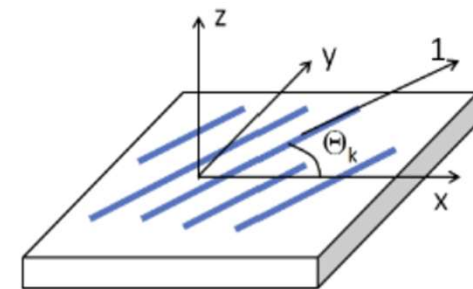
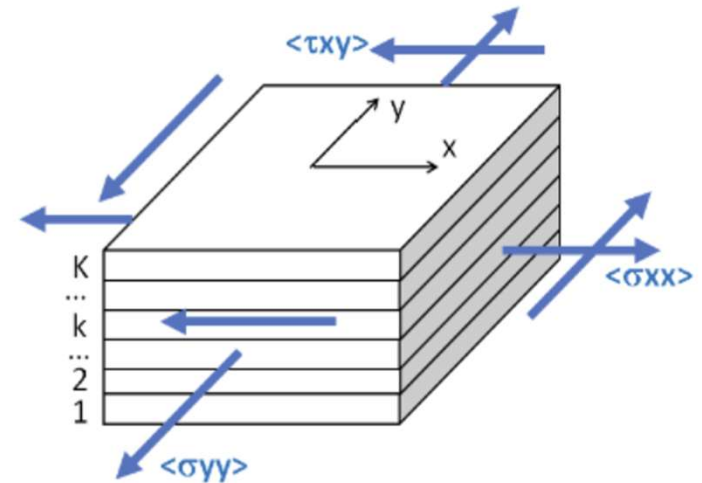
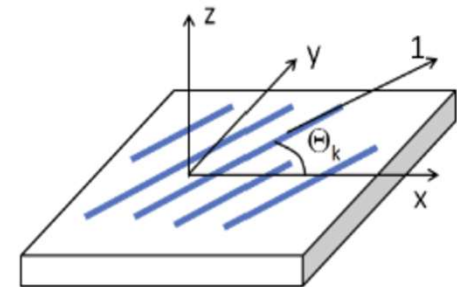
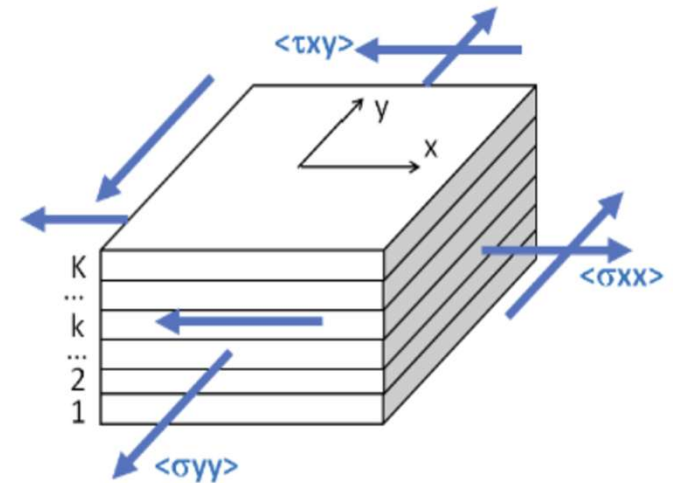


Table 6 Elastic Properties of Unidirectional Polymer Matrix Composites and 7075-T6 Aluminum

Fiber	Axial Modulus GPa	Transverse Modulus GPa	In-Plane Shear Modulus GPa	Major Poisson's Ratio
SM carbon (PAN)	145	10	4.1	0.25

ASSUMPTIONS

- **Classical Laminate Theory (CLT)** applies:
 - Plane stress condition ($\sigma_z = \tau_{xz} = \tau_{yz} = 0$)
 - Linear elastic behavior
 - Perfect bonding between plies
- Laminate is **symmetric** $\Rightarrow$ bending-extension coupling matrix
 - $[B] = 0$
- Each ply is orthotropic with the given properties
- Uniform strain through thickness (due to symmetry and in-plane loading)
- No thermal or moisture effects



ALGORITHMS OF ANALYSIS

1. Compliance and stiffness in material coordinates (1-2)
2. Transformation matrices for a ply at angle θ
3. Ply stiffness in global coordinates
4. Extensional stiffness matrix
5. Average stiffness and compliance
6. Engineering constants
7. Global strains
8. Ply strains and stresses in material coordinates

$$(1) \quad [S] = \begin{bmatrix} \frac{1}{E_{11}} & -\frac{\nu_{12}}{E_{11}} & 0 \\ -\frac{\nu_{12}}{E_{11}} & \frac{1}{E_{22}} & 0 \\ 0 & 0 & \frac{1}{G_{12}} \end{bmatrix}, \quad [C] = [S]^{-1}$$

$$(2) \quad [T_\sigma] = \begin{bmatrix} c^2 & s^2 & 2cs \\ s^2 & c^2 & -2cs \\ -cs & cs & c^2 - s^2 \end{bmatrix}, \quad [T_\varepsilon] = \begin{bmatrix} c^2 & s^2 & cs \\ s^2 & c^2 & -cs \\ -2cs & 2cs & c^2 - s^2 \end{bmatrix}$$

where $c = \cos \theta$, $s = \sin \theta$.

$$(3) \quad [C_x] = [T_\sigma]^{-1}[C][T_\varepsilon]$$

$$(4) \quad [A] = \sum_{k=1}^N [C_x]_k h$$

$$(5) \quad \langle C \rangle = \frac{[A]}{H}, \quad H = Nh, \quad \langle S \rangle = \langle C \rangle^{-1}$$

$$(6) \quad E_{xx} = \frac{1}{\langle S_{11} \rangle}, \quad E_{yy} = \frac{1}{\langle S_{22} \rangle}, \quad G_{xy} = \frac{1}{\langle S_{33} \rangle}$$

$$(7) \quad \nu_{xy} = -\frac{\langle S_{12} \rangle}{\langle S_{11} \rangle}, \quad \nu_{yx} = -\frac{\langle S_{12} \rangle}{\langle S_{22} \rangle}$$

$$\{\varepsilon\}_{\text{global}} = \langle S \rangle \{\sigma\}_{\text{global}}$$

$$(8) \quad \{\varepsilon\}_{12} = [T_\varepsilon] \{\varepsilon\}_{\text{global}}, \quad \{\sigma\}_{12} = [C] \{\varepsilon\}_{12}$$

```

===== RESULTS FOR Q = 13.3° =====
Effective laminate properties:
Exx = 157.707 GPa
Eyy = 13.071 GPa
Gxy = 39.357 GPa
vxy = 1.9456

Global mid-plane strains:
eps_xx = 6.382516e-03
eps_yy = 1.652291e-03
gamma_xy= 1.588037e-02

Ply stresses in material coords (sig11, sig22, tau12) [MPa]:
Ply 1 (θ = +90.0°): [2984.014  3.025  50.018]
Ply 2 (θ = -90.0°): [ 835.986  53.206  67.022]
Ply 3 (θ = +30.0°): [3740.742 -14.654  15.759]
Ply 4 (θ = -30.0°): [-502.801  84.482  49.350]
Ply 5 (θ = -30.0°): [-502.801  84.482  49.350]
Ply 6 (θ = +30.0°): [3740.742 -14.654  15.759]
Ply 7 (θ = -90.0°): [ 835.986  53.206  67.022]
Ply 8 (θ = +90.0°): [2984.014  3.025  50.018]

===== MAX/MIN OVER SWEEP (0° to 90°) =====
Max Exx = 196.70 GPa at Q = 0.0 deg
Min Exx = 47.63 GPa at Q = 38.0 deg
Max Eyy = 159.79 GPa at Q = 90.0 deg
Min Eyy = 9.43 GPa at Q = 29.0 deg
Max Gxy = 69.59 GPa at Q = 45.0 deg
Min Gxy = 32.17 GPa at Q = 0.0 deg
Max vxy = 2.2123 at Q = 23.0 deg
Min vxy = 0.1764 at Q = 90.0 deg

```

MATLAB – OUTPUT TERMINAL

Property	Value
E_{xx}	157.707 GPa
E_{yy}	13.071 GPa
G_{xy}	39.357 GPa
ν_{xy}	1.9456
ν_{yx}	0.1613*
* $\nu_{yx} = \nu_{xy} \frac{E_{yy}}{E_{xx}}$	

EFFECTIVE LAMINATE PROPERTIES (FOR Q=13.3°)

Strain	Value
ϵ_{xx}	6.3825×10^{-3}
ϵ_{yy}	1.6523×10^{-3}
γ_{xy}	1.5880×10^{-2}

GLOBAL MID-PLANE STRAINS
(FOR Q=13.3°)

Ply	Angle (°)	ε_{11}	ε_{22}	γ_{12}
1	+13.3	9.624×10^{-3}	-1.589×10^{-3}	1.220×10^{-2}
2	-13.3	2.573×10^{-3}	5.472×10^{-3}	1.635×10^{-2}
3	+30	1.207×10^{-2}	-4.001×10^{-3}	3.843×10^{-3}
4	-30	-1.654×10^{-3}	7.629×10^{-3}	1.204×10^{-2}
5	-30	-1.654×10^{-3}	7.629×10^{-3}	1.204×10^{-2}
6	+30	1.207×10^{-2}	-4.001×10^{-3}	3.843×10^{-3}
7	-13.3	2.573×10^{-3}	5.472×10^{-3}	1.635×10^{-2}
8	+13.3	9.624×10^{-3}	-1.589×10^{-3}	1.220×10^{-2}
<i>Strains computed from ply stresses using $[S]$ (consistency check)</i>				

PLY STRAINS IN MATERIAL COORDINATES (FOR Q=13.3°)

Strains computed from ply stresses using $[S]$ (consistency check)

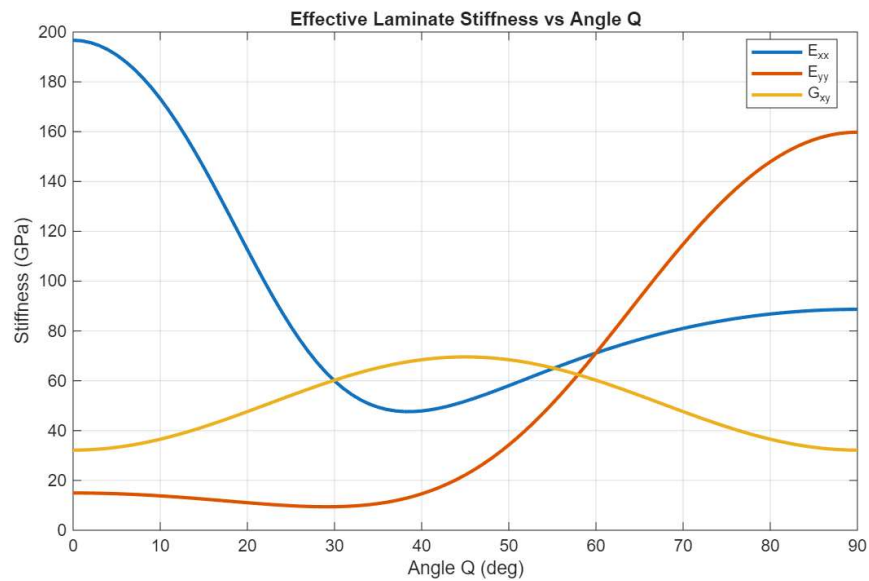
Ply	Angle (°)	σ_{11} (MPa)	σ_{22} (MPa)	τ_{12} (MPa)
1	+13.3	2984.014	3.025	50.018
2	-13.3	835.986	53.206	67.022
3	+30	3740.742	-14.654	15.759
4	-30	-502.801	84.482	49.350
5	-30	-502.801	84.482	49.350
6	+30	3740.742	-14.654	15.759
7	-13.3	835.986	53.206	67.022
8	+13.3	2984.014	3.025	50.018

PLY STRESSES IN MATERIAL COORDINATES
(FOR Q=13.3°)

VARIATION WITH Q– EFFECTIVE STIFFNESS

Observations:

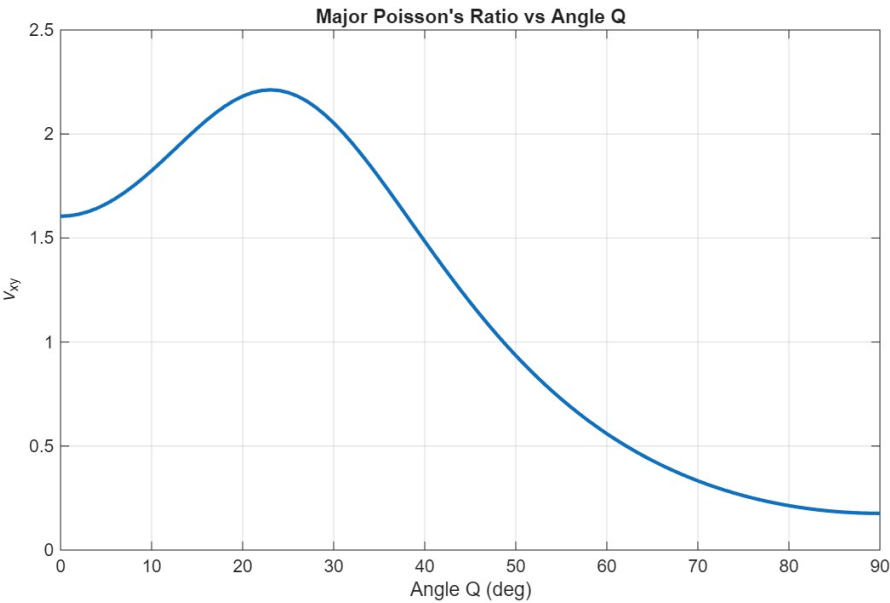
- E_{xx} dominated by plies near 0° , E_{yy} by plies near 90°
- G_{xy} peaks at 45° where shear coupling is maximum



Modulus	Maximum (GPa)	at Q ($^\circ$)	Minimum (GPa)	at Q ($^\circ$)
E_{xx}	196.70	0	47.63	38
E_{yy}	159.79	90	9.43	29
G_{xy}	69.59	45	32.17	0

VARIATION WITH Q– POISSON’S RATIO VXY

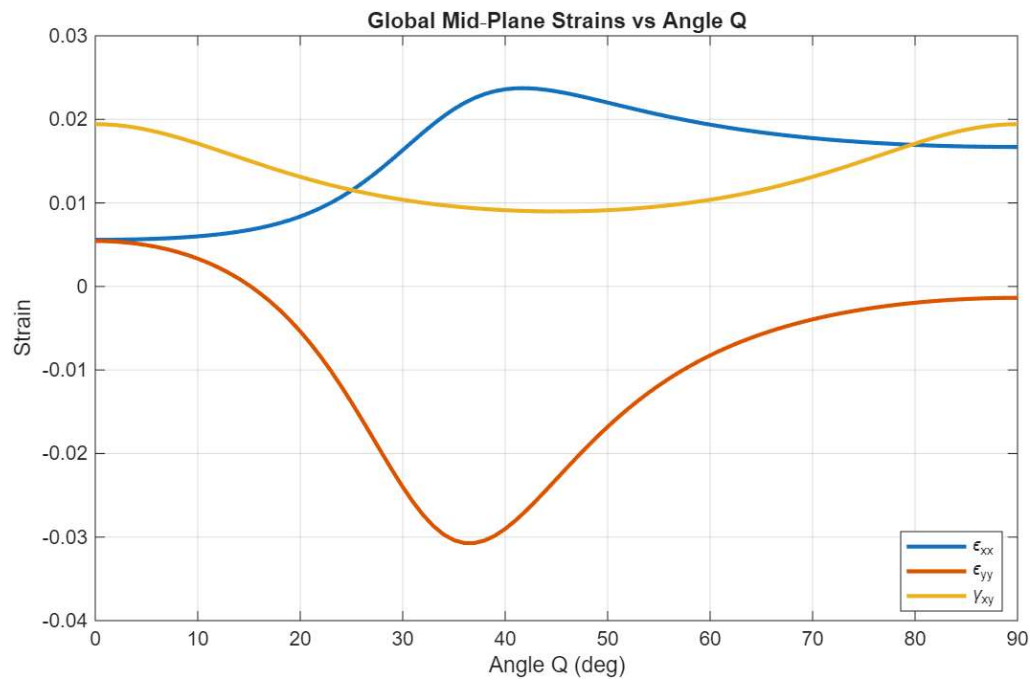
- Note: $\nu_{xy} > 1$ for a range of Q , which is possible in anisotropic laminates due to coupling



	Value	at Q (°)
Maximum ν_{xy}	2.2123	23
Minimum ν_{xy}	0.1764	90

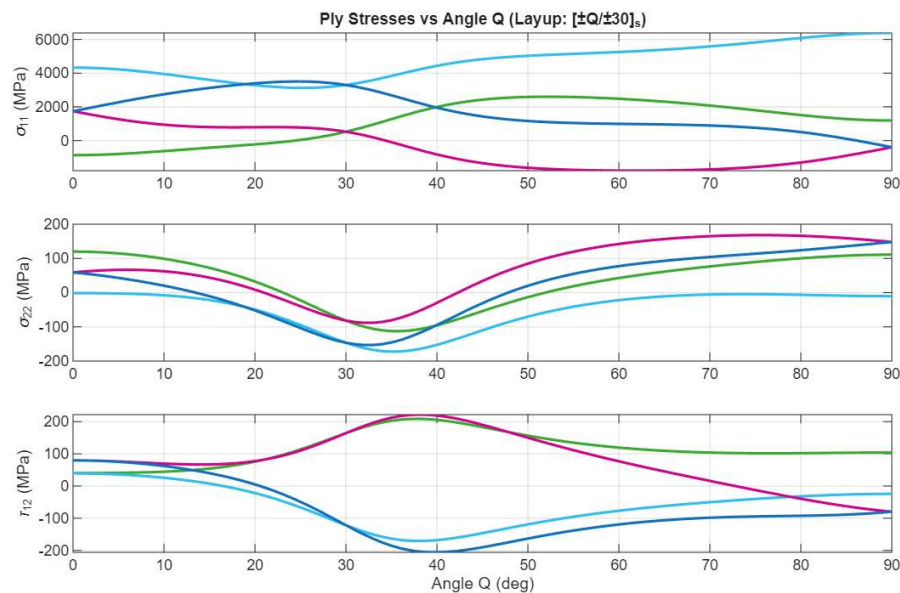
VARIATION WITH Q– GLOBAL STRAINS

- The strains vary significantly; extreme values occur at different Q



VARIATION WITH Q– PLY STRESSES

- The $\pm 30^\circ$ plies show very high tensile stresses (over 3700 MPa) near $Q = 13.3^\circ$, while the $\pm 13.3^\circ$ plies also experience high tension (≈ 3000 MPa)
- Compressive stresses appear in some plies (e.g., -30° plies show negative σ_{11})



CONCLUSIONS

- Classical Laminate Theory successfully applied to predict laminate behavior
 - For $Q = 13.3^\circ$:
 - $E_{xx} = 157.7 \text{ GPa}$ (stiff in x-direction)
 - $\nu_{xy} = 1.95$ indicates strong coupling
 - Global strains are on the order of 10^{-3} to 10^{-2}
 - **Ply stress levels:**
 - The $+30^\circ$ plies experience $\sigma_{11} = 3740 \text{ MPa}$, exceeding the SM carbon tensile strength (1520 MPa from Table 9).
 - The $+13.3^\circ$ plies also exceed strength (2984 MPa)
 - Therefore, **the laminate is likely to fail** under the given loads
 - Sweep analysis shows optimal angles for maximizing stiffness components, but strength must be considered
 - **Recommendation:** Redesign layup or reduce applied stresses to avoid failure
-

The code implements the CLT steps, sweeps over Q , and generates tables/plots.

es:

```

2 clear; tic; ticall;
3 % =====
4 % INPUT DATA (Dr Carbon/ply laminate with symmetric layup)
5 % =====
6 % Material Properties (fiber-aligned coordinates)
7 E11 = 38000; % [Pa] longitudinal modulus
8 E22 = 8200; % [Pa] transverse modulus
9 G12 = 4100; % [Pa] in-plane shear modulus
10 nu12 = 0.28; % major Poisson's ratio
11 ply_thick = 0.0001; % [m] ply thickness (used for Poisson, not essential)
12 % Applied global strains (given in MPa, converted to Pa)
13 eps_xx = 332000; % [1/m]
14 sigma_yy = 20800; % [Pa]
15 tau_xy = 42000; % [Pa]
16 global_strains = [sigma_xx; sigma_yy; tau_xy];
17 % Nonlinear angle  $\phi$  (Dr Carbon angle  $-11.3^\circ$ ) - used in layup, but we will use
18 % the value  $\phi = 11.3^\circ$  (in deg)
19 phi = 11.3; % [deg]
20 % COMPLIANCE AND STIFFNESS IN MATERIAL COORDINATES (1-2)
21 % =====
22 S = (1/E11 - nu12^2/E22) i1/i1;
23 S = (1/E22 - nu12^2/E11) i2/i2;
24 S = -nu12/E11 i1/i2;
25 C = 1/(1 - nu12^2) * S; % stiffness matrix in material axes
26 % =====
27 % SHEEP ORDER  $\phi$  TO STUDY INFLUENCE (0 to  $90^\circ$ )
28 % =====
29 Q_range = 0:1000; % [deg] sweep range
30 num_Q = length(Q_range);
31 % Preallocate arrays for storing results
32 Ex_val = zeros(1, num_Q);
33 Eyy_val = zeros(1, num_Q);
34 Gxy_val = zeros(1, num_Q);
35 vxy_val = zeros(1, num_Q);
36 ex_val = zeros(1, num_Q);
37 ey_val = zeros(1, num_Q);
38 gxy_val = zeros(1, num_Q);
39 % Ply stresses (8 plies, each column corresponds to one Q)
40 sig1_val = zeros(8, num_Q);
41 sig2_val = zeros(8, num_Q);
42 tau12_val = zeros(8, num_Q);
43 % =====
44 % Main loop
45 for id = 1:num_Q
46     Q = Q_range(id);
47     % Derive anisotropic layer:  $\cos(\phi)/\sin(\phi)$ 
48     % This gives 8 plies with  $\phi = 0^\circ, 15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ, 0^\circ$ 

```

[illegible]

```

175: plot(range, exp_yvals, 'linestyle', 2); hold on;
176: plot(range, exp_yrvals, 'linestyle', 2);
177: plot(range, gamma_yvals, 'linestyle', 2);
178: label('range');
179: label('exp');
180: label('gamma');
181: label('exp_r');
182: legend('linestyle', 'linestyle', 2);
183: legend('linestyle', 'linestyle', 2);
184:
185: % Figure 1: Effective stiffness model
186:
187: plot(range, ExpVals1, 'linestyle', 2); hold on;
188: plot(range, ExpVals2, 'linestyle', 2);
189: plot(range, ExpVals3, 'linestyle', 2);
190: label('range Q (deg)');
191: label('stiffness (N/m)');
192:
193: title('Effective Laminate Stiffness vs Angle Q');
194: legend('linestyle', 'linestyle', 2);
195: legend('linestyle', 'linestyle', 2);
196: grid on;
197:
198: % Figure 2: Poisson's ratio vxy
199:
200: plot(range, vxy_vals, 'linestyle', 2);
201: label('range (deg)');
202: label('vxy');
203:
204: title('Major Poisson's ratio vs Angle Q');
205: grid on;
206:
207: % =====
208:
209: Exp_max = max(ExpVals);
210: Exp_min = min(ExpVals);
211: Exp_r_max = max(ExpVals);
212: Exp_r_min = min(ExpVals);
213: vxy_max = max(vxy_vals);
214: vxy_min = min(vxy_vals);
215:
216: fprintf('===== RESULTS OVER SHEET (P = 98°) =====\n');
217: fprintf('Max E = %2.0f GPa @ %1.0f angle: Exp_max1, Q_range1_max Exp1\n');
218: fprintf('Min E = %2.0f GPa @ %1.0f angle: Exp_min1, Q_range1_min Exp1\n');
219: fprintf('Max E_r = %2.0f GPa @ %1.0f angle: Exp_r_max1, Q_range1_max Exp_r1\n');
220: fprintf('Min E_r = %2.0f GPa @ %1.0f angle: Exp_r_min1, Q_range1_min Exp_r1\n');
221: fprintf('Max vxy = %2.0f @ %1.0f angle: vxy_max, Q_range1_max vxy1\n');
222: fprintf('Min vxy = %2.0f @ %1.0f angle: vxy_min, Q_range1_min vxy1\n');
223:
224: fprintf('Max E = %2.0f GPa @ %1.0f angle: Exp_max2, Q_range2_max Exp2\n');
225: fprintf('Min E = %2.0f GPa @ %1.0f angle: Exp_min2, Q_range2_min Exp2\n');
226: fprintf('Max E_r = %2.0f GPa @ %1.0f angle: Exp_r_max2, Q_range2_max Exp_r2\n');
227: fprintf('Min E_r = %2.0f GPa @ %1.0f angle: Exp_r_min2, Q_range2_min Exp_r2\n');
228: fprintf('Max vxy = %2.0f @ %1.0f angle: vxy_max, Q_range2_max vxy2\n');
229: fprintf('Min vxy = %2.0f @ %1.0f angle: vxy_min, Q_range2_min vxy2\n');

```